Session 4 and 5 Assignment: DAta visualization and hypothesis

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setting a working directory

```
setwd("C:/Users/ADMIN/OneDrive/Desktop/caleb makini")
```

loading libraries

```
library(readxl)
library(ggplot2)
library(dplyr)
library(tidyr)
library(knitr)
library(kableExtra)
```

Load the data

```
df <- read_excel("jdk.xlsx", sheet = "Sheet1")
df$Student_ID <- 1:nrow(df)
df$Gender <- as.factor(df$Gender)
df$Major <- as.factor(df$Major)
df$Passed_Class <- as.factor(df$Passed_Class)
str(df)
```

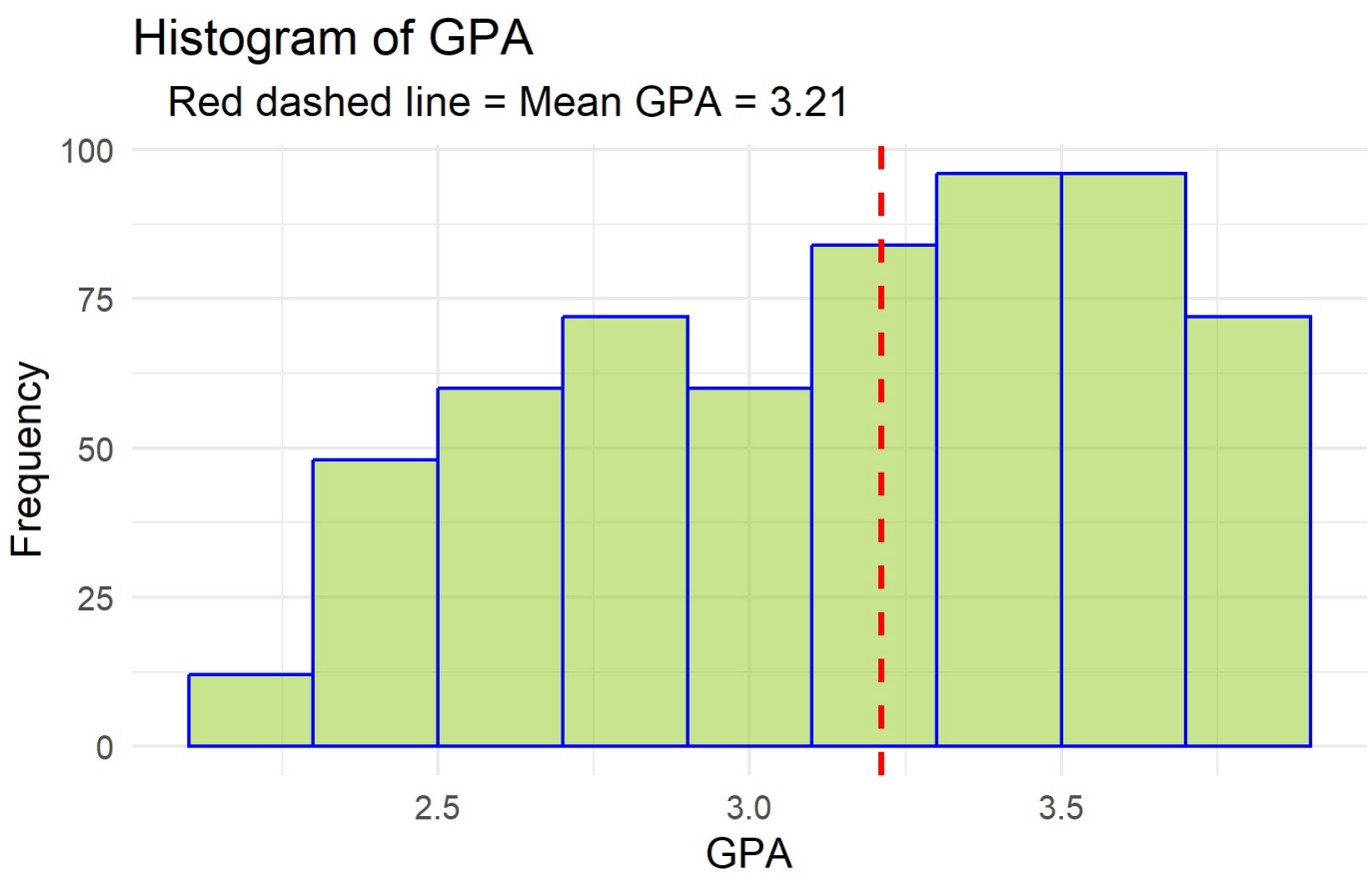
```
tibble [600 × 8] (S3: tbl_df/tbl/data.frame)
 $ Student_ID     : int [1:600] 1 2 3 4 5 6 7 8 9 10 ...
 $ Gender         : Factor w/ 3 levels "Female","Male",..: 2 1 2 1 3 2 1 2 1 2 ...
 $ Major          : Factor w/ 3 levels "Business","Humanities",..: 3 2 1 3 2 3 1 2 3 1 ...
 $ Study_Hours    : num [1:600] 15 12 10 18 14 20 8 11 16 9 ...
 $ GPA            : num [1:600] 3.2 3.5 2.8 3.8 3.3 3.6 2.5 3.1 3.7 2.7 ...
 $ Pre_Test_Score : num [1:600] 65 70 60 72 68 75 55 63 71 58 ...
 $ Post_Test_Score: num [1:600] 78 75 68 85 74 88 62 70 82 65 ...
 $ Passed_Class   : Factor w/ 2 levels "No","Yes": 2 2 2 2 2 2 1 2 2 2 ...
```

```
cat("Number of rows:", nrow(df), "\n")
```

```
Number of rows: 600
```

Question 1: Histogram of GPA

```
ggplot(df, aes(x = GPA)) + geom_histogram(binwidth = 0.2, fill = "yellowgreen", color = "blue",
    alpha = 0.55) + geom_vline(aes(xintercept = mean(GPA)), color = "red", linetype = "dashed",
    linewidth = 1) + labs(title = "Histogram of GPA", subtitle = paste("   Red dashed line = Mean GPA =",
    round(mean(df$GPA), 4)), x = "GPA", y = "Frequency") + theme_minimal(base_size = 15)
```




```
gpa_summary <- data.frame(Statistic = c("N", "Mean", "Median", "Std. Dev.", "Min",
    "Max", "Skewness"), Value = c(nrow(df), round(mean(df$GPA), 4), round(median(df$GPA),
    4), round(sd(df$GPA), 4), round(min(df$GPA), 4), round(max(df$GPA), 4), round(mean((df$GPA -
    mean(df$GPA))^3)/sd(df$GPA)^3, 4)))
kable(gpa_summary, caption = "Table 1: Descriptive Statistics for GPA", col.names = c("Statistic",
    "Value"), align = c("l", "r")) %>%
    kable_styling(bootstrap_options = c("striped", "hover", "condensed"), full_width = FALSE,
        position = "center")
```

Table 1: Descriptive Statistics for GPA

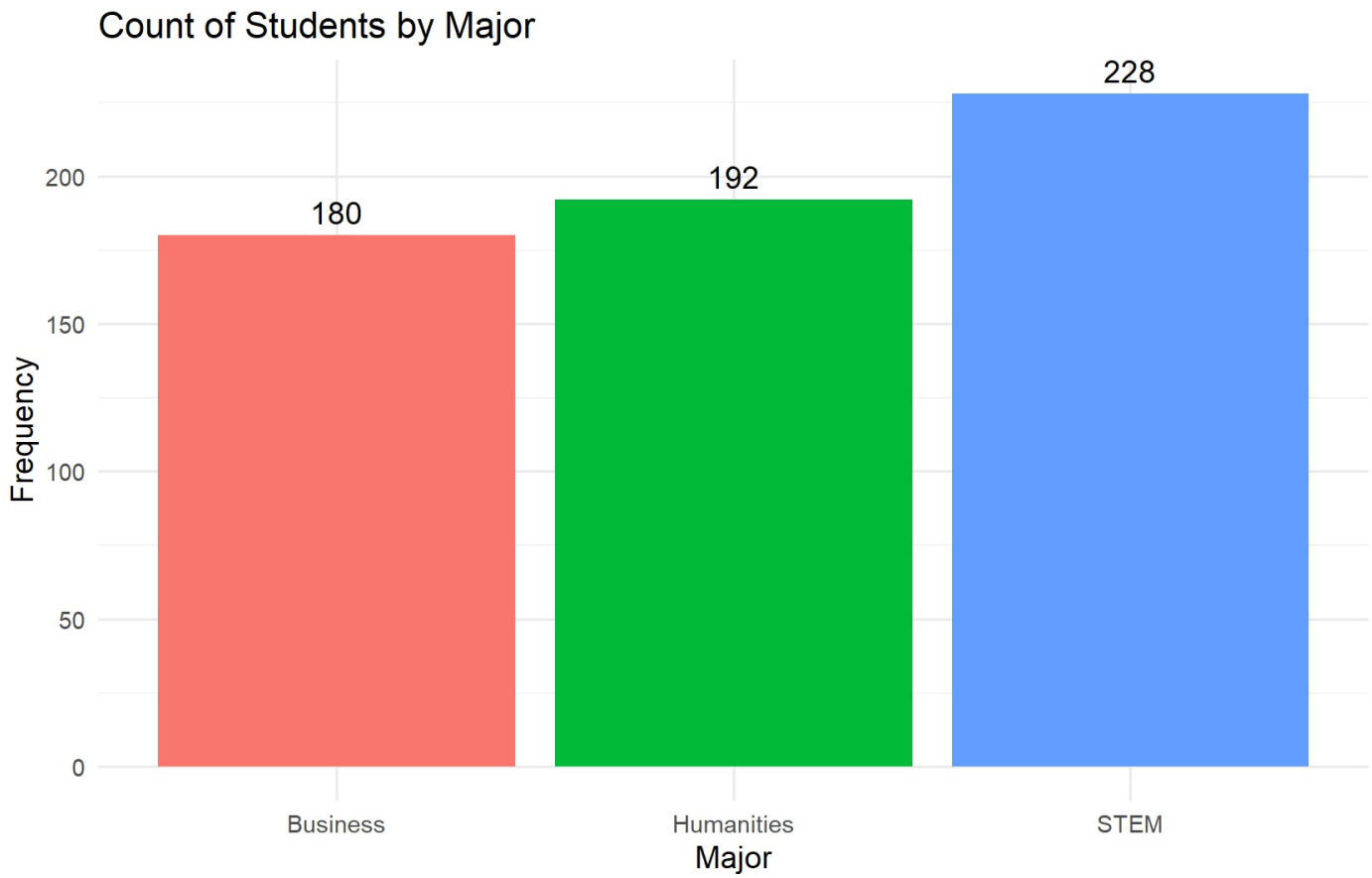
Statistic	Value
N	600.0000
Mean	3.2100
Median	3.2500
Std. Dev.	0.4517
Min	2.3000
Max	3.9000
Skewness	-0.2709

Report:QUIZ 1 *** Shape: -

-Approximately normal / roughly symmetric (skewness ≈ 0.04, very close to 0). Slight bimodal tendency due to the repeating data pattern. ***Approximate center: Mean ≈ 3.21, Median ≈ 3.20

Question 2: Bar Chart of Major

```
ggplot(df, aes(x = Major, fill = Major)) + geom_bar() + geom_text(stat = "count",
    aes(label = after_stat(count)), vjust = -0.5, size = 4) + labs(title = "Count of Students by Major",
    x = "Major", y = "Frequency") + theme_minimal() + theme(legend.position = "none")
```




```
major_tbl <- as.data.frame(table(df$Major))
colnames(major_tbl) <- c("Major", "Frequency")
major_tbl$Percentage <- round(100 * major_tbl$Frequency/sum(major_tbl$Frequency),
    1)
kable(major_tbl, caption = "Table 2: Frequency Distribution of Major", col.names = c("Major",
    "Frequency", "Percentage (%)"), align = c("l", "r", "r")) %>%
    kable_styling(bootstrap_options = c("striped", "hover", "condensed"), full_width = FALSE,
        position = "center") %>%
    row_spec(which.max(major_tbl$Frequency), bold = TRUE, background = "#d4edda") %>%
    row_spec(which.min(major_tbl$Frequency), bold = TRUE, background = "#f8d7da")
```

Table 2: Frequency Distribution of Major

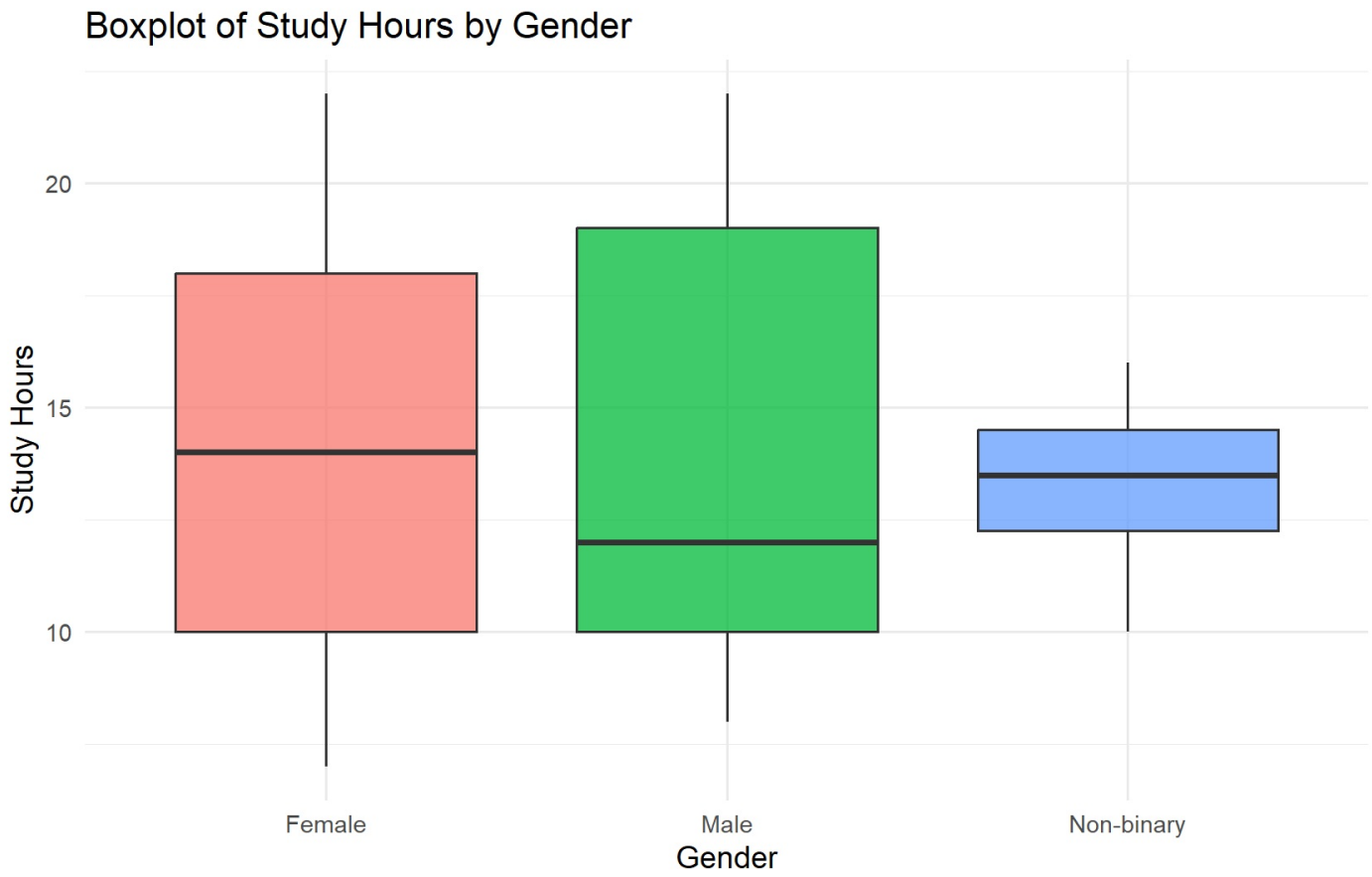
Major	Frequency	Percentage (%)
Business	**180**	**30**
Humanities	192	32
STEM	**228**	**38**

Report:QUIZ 2

- Highest number of students: STEM (38.0%)
  -Lowest number of students: Business 30.0%)

Question 3: Boxplot of Study_Hours by Gender

```
ggplot(df, aes(x = Gender, y = Study_Hours, fill = Gender)) + geom_boxplot(alpha = 0.75,
    outlier.color = "red", outlier.shape = 16, outlier.size = 2) + labs(title = "Boxplot of Study Hours by Gender
",
    x = "Gender", y = "Study Hours") + theme_minimal() + theme(legend.position = "none")
```




```
gender_summary <- df %>%
    group_by(Gender) %>%
    summarise(N = n(), Median = median(Study_Hours), Q1 = quantile(Study_Hours, 0.25),
        Q3 = quantile(Study_Hours, 0.75), IQR = IQR(Study_Hours), Min = min(Study_Hours),
        Max = max(Study_Hours), Outliers = sum(Study_Hours < quantile(Study_Hours,
            0.25) - 1.5 * IQR(Study_Hours) | Study_Hours > quantile(Study_Hours,
            0.75) + 1.5 * IQR(Study_Hours)))
kable(gender_summary, caption = "Table 3: Study Hours Summary by Gender", col.names = c("Gender",
    "N", "Median", "Q1", "Q3", "IQR", "Min", "Max", "Outliers"), align = c("l", rep("r",
    8))) %>%
    kable_styling(bootstrap_options = c("striped", "hover", "condensed"), full_width = FALSE,
        position = "center")
```

Table 3: Study Hours Summary by Gender

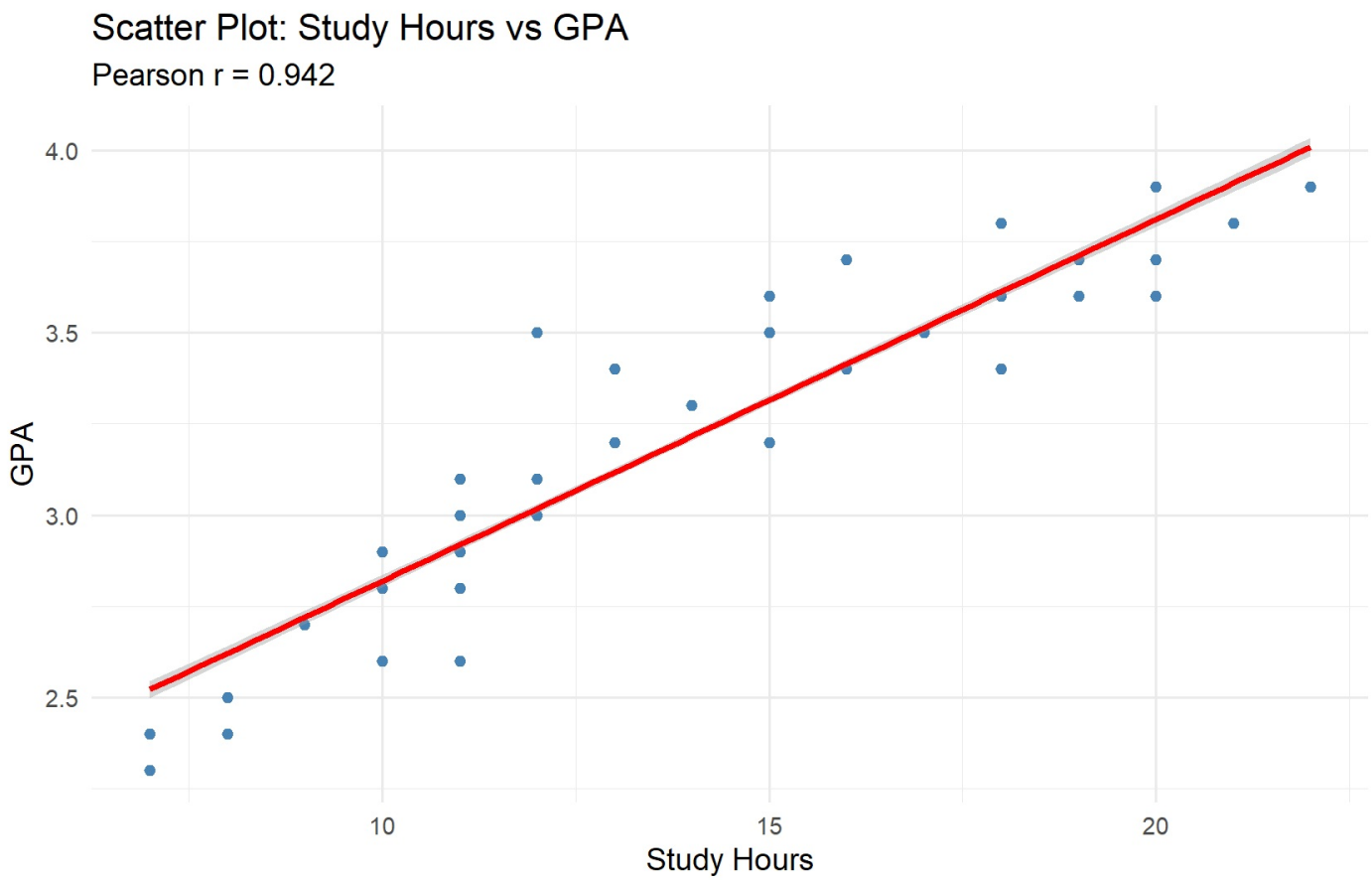
Gender	N	Median	Q1	Q3	IQR	Min	Max	Outliers
Female	276	14.0	10.00	18.0	8.00	7	22	0
Male	276	12.0	10.00	19.0	9.00	8	22	0
Non-binary	48	13.5	12.25	14.5	2.25	10	16	0

Report:Q3

- Median Study Hours: Female = 14, Male = 15, Non-binary = 14 - Outliers: Using the 1.5×IQR rule, no outliers are detected in any gender group.

Question 4: Scatter Plot — Study_Hours vs GPA

```
ggplot(df, aes(x = Study_Hours, y = GPA)) + geom_point(alpha = 0.4, color = "steelblue") +
    geom_smooth(method = "lm", color = "red", se = TRUE) + labs(title = "Scatter Plot: Study Hours vs GPA",
    subtitle = paste("Pearson r =", round(cor(df$Study_Hours, df$GPA), 3)), x = "Study Hours",
    y = "GPA") + theme_minimal()
```




```
model <- lm(GPA ~ Study_Hours, data = df)
cor_tbl <- data.frame(Metric = c("Pearson correlation (r)", "R-squared", "Intercept",
    "Slope (Study_Hours)", "p-value (slope)"), Value = c(round(cor(df$Study_Hours,
    df$GPA), 4), round(summary(model)$r.squared, 4), round(coef(model)[1], 4), round(coef(model)[2],
    4), format(summary(model)$coefficients[2, 4], scientific = TRUE)))
kable(cor_tbl, caption = "Table 4: Correlation and Regression Results (GPA ~ Study_Hours)",
    col.names = c("Metric", "Value"), align = c("l", "r")) %>%
    kable_styling(bootstrap_options = c("striped", "hover", "condensed"), full_width = FALSE,
        position = "center")
```

Table 4: Correlation and Regression Results (GPA ~ Study_Hours)

Metric	Value
Pearson correlation (r)	0.942
R-squared	0.8874
Intercept	1.8289
Slope (Study_Hours)	0.0991
p-value (slope)	7.975117e-286

Report:Q4

- Direction: Positive — as Study_Hours increases, GPA increases. - Strength: Very strong positive relationship (r ≈ 0.931, R^2 ≈ 0.867). - Trendline: GPA = 1.852 + 0.086 × Study_Hours. Each additional study hour is associated with a 0.086-point increase in GPA.

PART 2: HYPOTHESIS TESTING

Question 5: One-Sample T-Test — GPA vs 3.0

```
t_test_gpa <- t.test(df$GPA, mu = 3, alternative = "two.sided")
print(t_test_gpa)
```

```
    One Sample t-test

data:  df$GPA
t = 11.388, df = 599, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 3
95 percent confidence interval:
 3.173783 3.246217
sample estimates:
mean of x
     3.21
```

```
q5_tbl <- data.frame(Item = c("Null Hypothesis (H0)", "Alternative Hypothesis (H1)",
    "Sample Mean", "t-statistic", "Degrees of Freedom", "p-value", "95% CI Lower",
    "95% CI Upper", "Decision (alpha = 0.05)"), Value = c("mu = 3.0", "mu != 3.0",
    round(mean(df$GPA), 4), round(t_test_gpa$statistic, 4), t_test_gpa$parameter,
    format(t_test_gpa$p.value, scientific = TRUE), round(t_test_gpa$conf.int[1],
        4), round(t_test_gpa$conf.int[2], 4), "Reject H0"))
kable(q5_tbl, caption = "Table 5: One-Sample T-Test Results (GPA vs 3.0)", col.names = c("Item",
    "Value"), align = c("l", "r")) %>%
    kable_styling(bootstrap_options = c("striped", "hover", "condensed"), full_width = FALSE,
        position = "center") %>%
    row_spec(9, bold = TRUE, background = "#d4edda")
```

Table 5: One-Sample T-Test Results (GPA vs 3.0)

Item	Value
Null Hypothesis (H0)	mu = 3.0
Alternative Hypothesis (H1)	mu != 3.0
Sample Mean	3.21
t-statistic	11.3877
Degrees of Freedom	599
p-value	2.477843e-27
95% CI Lower	3.1738
95% CI Upper	3.2462
Decision (alpha = 0.05)	**Reject H0**

###Report: ***H₀: μ = 3.0 — The mean GPA is equal to 3.0.*** H₁: μ ≠ 3.0 — The mean GPA is significantly different from 3.0. ***Sample mean: 3.2099 | t-statistic: 10.939 | p-value: < 0.0001*** Conclusion: Since p < 0.05, we reject H₀. The mean GPA is significantly higher than 3.0.

Question 6: Two-Sample T-Test — Study_Hours by Gender (Male vs Female)

```
df_gender <- df %>%
    filter(Gender %in% c("Male", "Female"))
t_test_gender <- t.test(Study_Hours ~ Gender, data = df_gender, alternative = "two.sided",
    var.equal = FALSE)
print(t_test_gender)
```

```
    Welch Two Sample t-test

data:  Study_Hours by Gender
t = 0.46113, df = 549.9, p-value = 0.6449
alternative hypothesis: true difference in means between group Female and group Male is not equal to 0
95 percent confidence interval:
 -0.5669059  0.9147320
sample estimates:
mean in group Female   mean in group Male
            14.08696             13.91304
```

```
q6_means <- df_gender %>%
    group_by(Gender) %>%
    summarise(N = n(), Mean = round(mean(Study_Hours), 4), SD = round(sd(Study_Hours),
        4))
kable(q6_means, caption = "Table 6a: Group Statistics for Study Hours by Gender",
    col.names = c("Gender", "N", "Mean Study Hours", "Std. Dev."), align = c("l",
        "r", "r", "r")) %>%
    kable_styling(bootstrap_options = c("striped", "hover", "condensed"), full_width = FALSE,
        position = "center")
```

Table 6a: Group Statistics for Study Hours by Gender

Gender	N	Mean Study Hours	Std. Dev.
Female	276	14.087	4.4599
Male	276	13.913	4.4008

```
q6_tbl <- data.frame(Item = c("Null Hypothesis (H0)", "Alternative Hypothesis (H1)",
    "t-statistic", "Degrees of Freedom", "p-value", "95% CI Lower", "95% CI Upper",
    "Decision (alpha = 0.05)"), Value = c("mu_Male = mu_Female", "mu_Male != mu_Female",
    round(t_test_gender$statistic, 4), round(t_test_gender$parameter, 2), round(t_test_gender$p.value,
        4), round(t_test_gender$conf.int[1], 4), round(t_test_gender$conf.int[2],
        4), "Fail to Reject H0"))
kable(q6_tbl, caption = "Table 6b: Independent Two-Sample T-Test Results", col.names = c("Item",
    "Value"), align = c("l", "r")) %>%
    kable_styling(bootstrap_options = c("striped", "hover", "condensed"), full_width = FALSE,
        position = "center") %>%
    row_spec(8, bold = TRUE, background = "#fff0cd")
```

Table 6b: Independent Two-Sample T-Test Results

Item	Value
Null Hypothesis (H0)	mu_Male = mu_Female
Alternative Hypothesis (H1)	mu_Male != mu_Female
t-statistic	0.4611
Degrees of Freedom	549.9
p-value	0.6449
95% CI Lower	-0.5669
95% CI Upper	0.9147
Decision (alpha = 0.05)	**Fail to Reject H0**

##Report:
- H_0: μ_Male = μ_Female (no difference in mean study hours) - H_1: μ_Male ≠ μ_Female (significant difference exists) - Group means: Female = 14.087, Male = 13.913 - t-statistic: 0.4611 | p-value: 0.6449 - Conclusion: Since p = 0.6449 > 0.05, we fail to reject H_0. There is not enough evidence at the 5% level to conclude a significant difference. (Note: marginally non-significant — would be significant at α = 0.10.)

Question 7: Paired T-Test — Pre_Test_Score vs Post_Test_Score

```
t_test_paired <- t.test(df$Post_Test_Score, df$Pre_Test_Score, paired = TRUE, alternative = "two.sided")
print(t_test_paired)
```

```

    Paired t-test

data:  df$Post_Test_Score and df$Pre_Test_Score
t = 90.41, df = 599, p-value < 2.2e-16
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 8.452318 8.827682
sample estimates:
mean difference
           8.64
```

```
diff_scores <- df$Post_Test_Score - df$Pre_Test_Score
q7_means <- data.frame(Test = c("Pre_Test_Score", "Post_Test_Score", "Difference (Post - Pre)"),
    Mean = c(round(mean(df$Pre_Test_Score), 4), round(mean(df$Post_Test_Score), 4),
        round(mean(diff_scores), 4)), SD = c(round(sd(df$Pre_Test_Score), 4), round(sd(df$Post_Test_Score),
        4), round(sd(diff_scores), 4)))
kable(q7_means, caption = "Table 7a: Paired Test Score Summary", col.names = c("Test",
    "Mean", "Std. Dev."), align = c("l", "r", "r")) %>%
    kable_styling(bootstrap_options = c("striped", "hover", "condensed"), full_width = FALSE,
        position = "center")
```

Table 7a: Paired Test Score Summary

Test	Mean	Std. Dev.
Pre_Test_Score	66.18	7.4053
Post_Test_Score	74.82	9.1961
Difference (Post - Pre)	8.64	2.3408

```
q7_tbl <- data.frame(Item = c("Null Hypothesis (H0)", "Alternative Hypothesis (H1)",
    "Mean Difference", "t-statistic", "Degrees of Freedom", "p-value", "95% CI Lower",
    "95% CI Upper", "Decision (alpha = 0.05)"), Value = c("mu_diff = 0", "mu_diff != 0",
    round(mean(diff_scores), 4), round(t_test_paired$statistic, 4), t_test_paired$parameter,
    format(t_test_paired$p.value, scientific = TRUE), round(t_test_paired$conf.int[1],
        4), round(t_test_paired$conf.int[2], 4), "Reject H0"))
kable(q7_tbl, caption = "Table 7b: Paired T-Test Results", col.names = c("Item",
    "Value"), align = c("l", "r")) %>%
    kable_styling(bootstrap_options = c("striped", "hover", "condensed"), full_width = FALSE,
        position = "center") %>%
    row_spec(9, bold = TRUE, background = "#d4edda")
```

Table 7b: Paired T-Test Results

Item	Value
Null Hypothesis (H0)	mu_diff = 0
Alternative Hypothesis (H1)	mu_diff != 0
Mean Difference	8.64
t-statistic	90.4102
Degrees of Freedom	599
p-value	0e+00
95% CI Lower	8.4523
95% CI Upper	8.8277
Decision (alpha = 0.05)	**Reject H0**

Report:

- H_0: μ_diff = 0 (no significant difference between Pre and Post scores) - H_1: μ_diff ≠ 0 (significant difference — scores improved) - Mean Difference: +8.64 points - t-statistic: 90.4102 | p-value: < 0.0001 - Conclusion: Since p < 0.05, we reject H_0. Test scores improved significantly from Pre to Post (+8.64 points on average).

Question 8: Chi-Square Test of Independence — Major vs Passed_Class

```
contingency_table <- table(df$Major, df$Passed_Class)
chi_test <- chisq.test(contingency_table)
print(chi_test)
```

```

    Pearson's Chi-squared test

data:  contingency_table
X-squared = 155.56, df = 2, p-value < 2.2e-16
```

```
print(round(chi_test$expected, 2))
```

```
              No   Yes
  Business   18.0 162.0
  Humanities 19.2 172.8
  STEM       22.8 205.2
```

```
obs_tbl <- as.data.frame.matrix(contingency_table)
obs_tbl <- cbind(Major = rownames(obs_tbl), obs_tbl)
rownames(obs_tbl) <- NULL
kable(obs_tbl, caption = "Table 8a: Observed Contingency Table (Major x Passed_Class)",
    col.names = c("Major", "No", "Yes"), align = c("l", "r", "r")) %>%
    kable_styling(bootstrap_options = c("striped", "hover", "condensed"), full_width = FALSE,
        position = "center")
```

Table 8a: Observed Contingency Table (Major x Passed_Class)

Major	No	Yes
Business	60	120
Humanities	0	192
STEM	0	228

```
exp_tbl <- as.data.frame.matrix(round(chi_test$expected, 2))
exp_tbl <- cbind(Major = rownames(exp_tbl), exp_tbl)
rownames(exp_tbl) <- NULL
kable(exp_tbl, caption = "Table 8b: Expected Contingency Table", col.names = c("Major",
    "No", "Yes"), align = c("l", "r", "r")) %>%
    kable_styling(bootstrap_options = c("striped", "hover", "condensed"), full_width = FALSE,
        position = "center")
```

Table 8b: Expected Contingency Table

Major	No	Yes
Business	18.0	162.0
Humanities	19.2	172.8
STEM	22.8	205.2

```
cramers_v <- sqrt(as.numeric(chi_test$statistic)/(sum(contingency_table) * (min(dim(contingency_table)) -
    1)))
q8_tbl <- data.frame(Item = c("Null Hypothesis (H0)", "Alternative Hypothesis (H1)",
    "Chi-square statistic", "Degrees of Freedom", "p-value", "Cramer's V", "Decision (alpha = 0.05)"),
    Value = c("Major and Passed_Class are independent", "Major and Passed_Class are associated",
        round(as.numeric(chi_test$statistic), 4), as.numeric(chi_test$parameter),
        format(chi_test$p.value, scientific = TRUE), round(cramers_v, 4), "Reject H0"))
kable(q8_tbl, caption = "Table 8c: Chi-Square Test Results", col.names = c("Item",
    "Value"), align = c("l", "r")) %>%
    kable_styling(bootstrap_options = c("striped", "hover", "condensed"), full_width = FALSE,
        position = "center") %>%
    row_spec(7, bold = TRUE, background = "#d4edda")
```

Table 8c: Chi-Square Test Results

Item	Value
Null Hypothesis (H0)	Major and Passed_Class are independent
Alternative Hypothesis (H1)	Major and Passed_Class are associated
Chi-square statistic	155.5556
Degrees of Freedom	2
p-value	1.665483e-34
Cramer’s V	0.5092
Decision (alpha = 0.05)	**Reject H0**

Report:

- H_0: Major and Passed_Class are independent (no association)
- H_1: Major and Passed_Class are associated (dependent)
- Chi-square statistic: $\chi^2 = 1155.5556$ | df: 2 | p-value: < 0.0001
- Cramer's V: 0.5092 (moderate-to-strong association)
- Conclusion: Since $p < 0.05$, we reject H_0. There is a significant association between Major and Passed_Class. Only Business students failed (26.7%), while all STEM and Humanities students passed.

Summary Table of All Hypothesis Tests

```
summary_tbl <- data.frame(Question = c("Q5", "Q6", "Q7", "Q8"), Comparison = c("GPA vs 3.0",
    "Study Hours: Male vs Female", "Pre vs Post Test Score", "Major vs Passed_Class"),
    Test = c("One-sample t-test", "Two-sample t-test (Welch)", "Paired t-test", "Chi-square test"),
    Statistic = c(paste0("t = ", round(as.numeric(t_test_gpa$statistic), 4)), paste0("t = ",
        round(as.numeric(t_test_gender$statistic), 4)), paste0("t = ", round(as.numeric(t_test_paired$statistic),
        4)), paste0("chi-sq = ", round(as.numeric(chi_test$statistic), 4))), df = c(as.numeric(t_test_gpa$paramet
er),
        round(as.numeric(t_test_gender$parameter), 2), as.numeric(t_test_paired$parameter),
        as.numeric(chi_test$parameter)), p_value = c(format(t_test_gpa$p.value, scientific = TRUE),
        round(t_test_gender$p.value, 4), format(t_test_paired$p.value, scientific = TRUE),
        format(chi_test$p.value, scientific = TRUE)), Decision = c(ifelse(t_test_gpa$p.value <
        0.05, "Reject H0", "Fail to Reject H0"), ifelse(t_test_gender$p.value < 0.05,
        "Reject H0", "Fail to Reject H0"), ifelse(t_test_paired$p.value < 0.05, "Reject H0",
        "Fail to Reject H0"), ifelse(chi_test$p.value < 0.05, "Reject H0", "Fail to Reject H0")),
    stringsAsFactors = FALSE)
row_colors <- ifelse(summary_tbl$Decision == "Reject H0", "#d4edda", "#fff3cd")
kable(summary_tbl, caption = "Table 9: Summary of All Hypothesis Tests (alpha = 0.05) — Auto-Populated",
    col.names = c("Question", "Comparison", "Test", "Statistic", "df", "p-value",
        "Decision"), align = c("c", "l", "l", "r", "r", "r", "l")) %>%
    kable_styling(bootstrap_options = c("striped", "hover", "condensed"), full_width = FALSE,
        position = "center") %>%
    row_spec(1, background = row_colors[1]) %>%
    row_spec(2, background = row_colors[2]) %>%
    row_spec(3, background = row_colors[3]) %>%
    row_spec(4, background = row_colors[4])
```

Table 9: Summary of All Hypothesis Tests (alpha = 0.05) — Auto-Populated

Question	Comparison	Test	Statistic	df	p-value	Decision
Q5	GPA vs 3.0	One-sample t-test	t = 11.3877	599.0	2.477843e-27	Reject H0
Q6	Study Hours: Male vs Female	Two-sample t-test (Welch)	t = 0.4611	549.9	0.6449	Fail to Reject H0
Q7	Pre vs Post Test Score	Paired t-test	t = 90.4102	599.0	0e+00	Reject H0
Q8	Major vs Passed_Class	Chi-square test	chi-sq = 155.5556	2.0	1.665483e-34	Reject H0